Product Specification Document

Indian Personal Finance & EMI Tracker App

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1. Executive Summary

This document defines the complete product specifications, technical architecture, database schemas, and regulatory compliance requirements for a cross-platform personal finance management application designed specifically for the Indian financial ecosystem. The system serves across Android, iOS, Web, and Windows Desktop, delivering automated expense tracking, smart budget alerts, credit health monitoring, and complete visibility over active equated monthly installments (EMIs).

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2. Target Platforms & Cross-Platform Architecture

To deliver a uniform experience while reducing time-to-market across mobile, desktop, and web environments, the application leverages a unified frontend paired with a high-concurrency microservice backend architecture.

Layer	Primary Selection	Secondary / Fallback Option	Core Architectural Purpose
Mobile, Web & Desktop	Flutter (Dart)	React Native + Next.js + ...	Single codebase rendering native UI on Android, iOS, Web, and Windows Desktop.
Backend Infrastructure	Node.js (TypeScript)	Python (FastAPI)	Asynchronous API layer handling webhook ingestions, ingestion triggers, and authentication.

Layer	Primary Selection	Secondary / Fallback Option	Core Architectural Purpose
Database Engine	PostgreSQL	—	Relational store enforcing strong consistency for transactions, balance tracking, and schema stability.
Caching & Queue Layer	Redis	—	High-performance in-memory cache for fast dashboard aggregation and rate-limiting SMS/AA triggers.

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3. Data Ingestion Channels (India Specific)

The system relies on three distinct data collection pipelines tailored specifically to the Indian financial ecosystem:

- **Account Aggregator (AA) Framework:** Connects directly to RBI-licensed Account Aggregators (e.g., Setu, Finvu, Anumati) via encrypted consent flow. Securely pulls verified bank statements, fixed deposit records, mutual fund investments, and real-time ledger balances.
- **On-Device Local SMS Parser:** Operates exclusively on local device memory (Android native runtime) to process transactional SMS alerts from banks and payment gateways (UPI, credit card charges, ATM debits). Ensures user data privacy by processing raw string patterns locally without cloud upload.
- **Credit Bureau Integration:** Periodically fetches formal credit profiles, active line limits, outstanding balances, and credit scores through Experian or CRIF High Mark APIs using PAN verification and OTP authentication.

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4. Core Features & User Interface Sections

4.1 Unified Net Standings Dashboard

- **Real-time Net Worth Calculation:** Displays immediate position using `Total Bank Balances - Total Credit Card Outstanding`.
- **Asset Breakdown:** Categorized view of savings accounts, fixed deposits, investments, and wallet balances.
- **Liability Summary:** Aggregate display of active loans, outstanding card balances, and upcoming obligations.

4.2 Monthly Utilization Tracker

- **Spend Velocity Gauge:** Visual progress indicators tracking total spend relative to user-defined monthly limits.
- **Categorization Engine:** Automated classification into spending verticals (Food, Utilities, Travel, Shopping, Health).
- **Payment Mode Analytics:** Breakdown by method: **UPI vs. Credit Card vs. Net Banking vs. ATM cash withdrawals**.
- **Risk Mitigation Alerts:** Automated push notification triggered instantly when total credit card credit utilization crosses the critical **30% threshold**.

4.3 EMI & Debt Command Center

- **Active Loan Overview:** Detailed registry of active credit card split-EMIs, personal loans, home loans, and buy-now-pay-later (BNPL) schemes.
- **Repayment Metrics:** Real-time visibility into remaining principal, monthly installment values, total interest costs, and remaining tenure.
- **Auto-Debit Calendar:** Visual schedule detailing payment due dates, bank mandate statuses, and automated reminders 3 days prior to debit events.

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5. PostgreSQL Database Schema

```
-- User Profile

CREATE TABLE users (

    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
    full_name VARCHAR(100) NOT NULL,
    phone_number VARCHAR(15) UNIQUE NOT NULL,
    pan_number VARCHAR(10) UNIQUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
-- Financial Accounts
CREATE TABLE accounts (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID REFERENCES users(id) ON DELETE CASCADE,
    account_type VARCHAR(20) CHECK (account_type IN ('SAVINGS', 'CURRENT',
'CREDIT_CARD', 'WALLET')),
    institution_name VARCHAR(50) NOT NULL,
    account_mask VARCHAR(4) NOT NULL,
    credit_limit DECIMAL(12,2) DEFAULT 0.00,
    current_balance DECIMAL(12,2) DEFAULT 0.00,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
-- Transactions
```

```
CREATE TABLE transactions (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID REFERENCES users(id) ON DELETE CASCADE,
    account_id UUID REFERENCES accounts(id) ON DELETE CASCADE,
    amount DECIMAL(12,2) NOT NULL,
    transaction_type VARCHAR(10) CHECK (transaction_type IN ('DEBIT', 'CREDIT')),
    payment_mode VARCHAR(20) CHECK (payment_mode IN ('UPI', 'CREDIT_CARD',
'DEBIT_CARD', 'NET_BANKING', 'ATM')),
    merchant_name VARCHAR(100),
    category VARCHAR(50),
    transaction_date TIMESTAMP NOT NULL,
    raw_sms_hash VARCHAR(64) UNIQUE
);
-- Active EMIs & Loans
CREATE TABLE active_emis (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID REFERENCES users(id) ON DELETE CASCADE,
    account_id UUID REFERENCES accounts(id) ON DELETE CASCADE,
    loan_title VARCHAR(100) NOT NULL,
```

```
    total_principal DECIMAL(12,2) NOT NULL,
    monthly_emi_amount DECIMAL(12,2) NOT NULL,
    total_tenure_months INT NOT NULL,
    remaining_tenure_months INT NOT NULL,
    emi_due_date INT CHECK (emi_due_date BETWEEN 1 AND 31),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

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6. Regulatory & Security Compliance

- **Digital Personal Data Protection (DPDP) Act Compliance:** Strictly implements user consent mechanisms prior to initiating data ingest pipelines. Features a dedicated privacy portal giving users full control to export data logs or trigger one-click profile purges.
- **Hardware-Backed Biometric Security:** Integrates directly with platform-native authentication APIs (Windows Hello, Android Fingerprint/Face Unlock, iOS Touch ID/Face ID) to guard application access and sensitive data rendering.
- **On-Device Data Isolation Architecture:** Unparsed transactional SMS strings are isolated within local sandbox memory. Regex parsing logic executes strictly on-device, ensuring raw personal messages are never transmitted to cloud environments.